

PHM — Study 1

Continuous damage estimation
Provisional image-derived target
20 run-level units
MAE / RMSE / Spearman



REASSEMBLE — Study 2

Execution-failure classification
4,530 segments / 148 recordings
AUROC / AUPRC / calibration
Demonstrated complementarity



Shared methodology

Validate targets and signal
Group-disjoint evaluation
Start with simple fusion
Test degraded modalities